

Quote precisely theorem/result from class you are using.
if using HA problem, prove it and use. Justify your steps. Total marks: 30.

1. $X \sim \text{Uniform}(a, b)$ with $E(X) = 3$ and $\text{variance}(X) = 3$.
 - (a) Find $P(3 < X < 7 | -1 < X < 5)$.
 - (b) Let Y be the area of square constructed with side length $|X - 1|$.
Find pdf of Y . [3+8]
2. Show that there is a tournament on n players which has at least $n!2^{-(n-1)}$ Hamiltonian paths. A Hamiltonian path is an ordering of the vertices $v_1, v_2, \dots, v_n$ such that for each $i < n$, $\overrightarrow{v_i v_{i+1}}$ is an edge. [4]
3. Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with variance σ^2 . Let $\bar{X} = (X_1 + X_2 + \dots + X_n)/n$, their average. Show that [7]

$$E \left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right] = \sigma^2.$$

4. (a) I toss a fair coin 50 times and X is the number of Heads obtained. you toss the fair coin 50 times and Y is the number of Tails you obtain. Calculate expectation of $(X - Y)(X + Y)$.
- (b) A coin, chance of Heads p , is tossed 10 times. Let X be the number of Heads and Y number of tails obtained. Calculate $E(XY)$
- (c) A company guarantees machines to be trouble free for the first five weeks after purchase. It is known that the time to first failure has exponential density with mean 10 weeks. One such machine is placed in service in the hostel. Upon inspection exactly 8 weeks later it is found to have failed at some time earlier. Given this, what is the probability that it actually failed in the first five weeks?
- (d) Two subsets X, Y of $\{1, 2, \dots, n\}$ are selected independently and at random from among all subsets. Find $P(X \subset Y)$. [2+2+2+2]

GOODLUCK